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Article

Adaptive AI Framework for Real-Time Threat Detection in IoT Ecosystems

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Abstract: Internet of Things (IoT) ecosystems have seen rapid growth, and this growth has led to a significantly broader cyber attack surface to which the resource-constrained edge devices are targeted. The traditional centralized detection systems result in high bandwidth, long transmission delay and privacy concerns. Deplotting Machine Learning models at the network edge is the solution to dealing with latency; static edge model quickly gets stale from concept drift and can have challenges in detecting zero-day exploits. In addition, Fine Tune Edge models dynamically often results in catastrophic forgetting, leading to a high False Positive Rate (FPR).

In this paper, we propose a lightweight Adaptive AI Framework for Real-Time Threat Detection in IoT Ecosystems in order to overcome such challenges. The framework is deployed at the edge layer, and runs with an 8-bit quantized neural engine as well as an Elastic Weight Consolidation (EWC) based Continual Learning (CL) mechanism. This architecture provides a low latency processing of network streams, realtime update of security models and allows maintenance of previous "benign" baselines without dependence on cloud.

Empirical experiments conducted on benchmark datasets namely N-BaIoT and Edge-IIoTset with an ARM-based edge gateway shows that the proposed framework achieves an overall detection accuracy of 98.88%, F1 score of 98.75% and False Positive Rate of 1.20%. Critically, has very low memory usage (<16 MB) and low to average latency on the average (2.95 ms) on every compute. The results shows that the framework is effective, highly adaptable and of great scalability in the context of resource constrained IoT application environment.

The main use cases covered include: IoT Security, Edge AI, Continual Learning, Elastic Weight Consolidation, Threat Detection, Low Latency, Concept Drift.

Keywords: IoT Security, Edge AI, Continual Learning, Elastic Weight Consolidation, Threat Detection

1. Introduction

1.1 Background and Context

Billions of resource-challenged sensors, actuators, and embedded edge devices now connect over industrial networks, smart infrastructure and healthcare systems, with a growth curve that appears to be exponential, known as the IoT ecosystem. This huge connectivity allows up to the-minute security monitoring and automation — but it also vastly increases the cyber threat attack surface available. Attackers have been increasingly targeting edge networks with sophisticated attacks, including Distributed Denial of Service (DDoS) botnets, like Mirai and BASHLITE, and zero day vulnerability exploits[1].

The old school security architectures are dependent on security information and event management (SIEM) running in the cloud or at a remote location for a threat monitor system. But, taking telemetry streams from the edge of the network to remote cloud servers creates major operational challenges:

- **Lack of Threat Identification (TI): Unacceptable delays that make instant automated mitigation difficult or impossible.**
- **Bandwidth Saturation:** Bursty bandwidth usage due to transmitting lots of raw, high volume of sensor network data.
- **Privacy Risks:** Greater risk of interception happening while raw data is in transport over wide area networks.

1.2 Problem Statement

The research shows that opting for Machine Learning (ML) threat detection models at the network edge (e.g., edge gateway and local microcontroller) can help to reduce latency and network overhead. There are however, three stringent academic and practical challenges presented by implementing real-time threat detection in IoT edge environments:

1. High accuracy Deep Learning models demand a large amount of memory (RAM) and computational power, which are higher than those of embedded IoT devices with strict processing and hardware resource constraints (≤16MB RAM).
2. When Network traffic patterns and Cyber attack vectors are dynamic, Static Model Degradation (SMD) arises. Performance of static detection models hungers over time because the models are fooled by new 'zero day' attacks (also known as ‘concept drift’) that occur at the edge.
3. In the dynamic learning setting the constant retraining of detection models on top of new attack patterns often results in a phenomenon called catastrophic forgetting causing the detection model to stop preserving learned network baselines (i.e., false negatives) for benign network traffic that causes high False Positive Rates (FPR) which are not acceptable.

1.3 Proposed Solution and Research Contributions

This paper addresses these issues by proposing an Adaptive AI Framework designed for Real-Time Threat Detection in IoT Ecosystems. The suggested framework is based on a lightweight quantized neural detector, run natively at the edge layer, and an adaptive Continual Learning (CL) engine running with an Elastic Weight Consolidation (EWC) algorithm. The architecture allows for real-time detection of threats without the dependence on a cloud system, without memory accumulation and without giving up to previous security baselines[2].

This work has brought the following main benefits:

- **Lightweight Edge-Native Pipeline**: A highly optimized feature extraction and quantized detection engine, optimized for use when machine requirements are constrained (≤3.5 ms latency, < 16 MB RAM).
- **Continual Learning without Catastrophical Forgetting**: Integration of an EWC-based dynamic update mechanism to learn local baseline benign traffic signatures and update along local emerging zero day threats continuously.
- **Empirical Validation on Benchmark Datasets**: Scrutineous tests on real-world IoT datasets (N-BaIoT and Edge-IIoTset) with an excellent detection accuracy of 98.88% and low false-alarm rate of 1.20% and negligible memory overhead.

1.4 Threat Detection Paradigms in IoT and Edge Ecosystems

Internet of Things (IoT) security architecture has undergone tremendous changes in recent ten years. Early solutions were mostly based on centralized cloud-based solutions, Security Information and Event Management (SIEM) and signature-based Intrusion

Detection Systems (IDS). Recognizable for fixed and known attack signatures and low apparent end-to-end transmission delay (e.g., < 200ms), cloud-centric paradigms require significant bandwidth overhead with large streams of raw IoT telemetry data and latency that is too high for an application that is intended to interact with users in real time[3].

Recent paradigms aim to move the workloads from the central data center closer to the edge, mitigating latency issues. It has been proposed to run classical lightweight Machine Learning (ML) models (Support Vector Machines (SVM), Random Forests (RF), and Naive Bayes) directly on the edge gateways. While such edge-architected models perform well in lowering processing latency, their static design makes them naturally susceptible to hackers with multi-stage, multi-pronged zero-day threats and changing attack vectors[4].

1.4.1 Concept Drift and Continual Learning at the Edge

One key challenge with edge-based cybersecurity is the dynamic nature of network traffic and the Concept Drift, whereby traffic behavior is not stationary over time and benign behavior and traffic patterns evolve. The recognition rate drops quickly and the False Positive Rate (FPR) is high when static ML models are stimulated to concept drift.

The traditional method of coping with concept drift has been to return edge collected data to a central server for periodical offline retraining. Yet, this method can pose two significant challenges in terms of operation in resource-constrained IoT environments :

1. **Bandwidth and Energy Exhaustion:** Telemetry data constantly floods low-power edge sensors and links all wires into wireless local channels, which soon exhausts the energy available in the battery.
2. **Catastrophic Forgetting:** When building a new neural network and finetuning it only on new attack samples collected without structural protections to prevent it, the model will be forced to forget its previous knowledge. This can cause loss of good network baselines, leading to significant alert fatigue.

However, Continuous Learning (CL) mechanisms, in particular regularized solutions such as Elastic Weight Consolidation (EWC), have proven to be a mathematically-valid approach, by discarding huge historical storage. EWC computes network parameters' importance levels, according to previous work, enabling incremental weight updates in case of new threats, meanwhile safeguarding critical memory parameters[5].

1.4.2 Comparative Analysis of Existing Frameworks

In order to demonstrate the impacts of this study, the following main methodologies in IoT threat detection were analyzed accordingly for the following factors: system architecture, detection time, memory consumption and dynamic adjustment.

Table 1. Comparative Taxonomy of IoT Threat Detection Frameworks vs. Proposed Adaptive Edge AI Framework

Research Study / Baseline Model	Core Algorithm	Deployment Layer	Avg Latency (ms)	False Positive Rate (%)	Memory Footprint (RAM)	Dynamic Continual Learning	Resistance to Catastrophic Forgetting
Al-Hawawreh et al. (2021)	Deep Autoencoders	Centralized Cloud	185.0 ms	4.80%	High (> 2 GB)	No	No
Zarpelão et al. (2020)	Static Random Forest	Edge Gateway	24.5 ms	5.10%	Moderate (~ 256 MB)	No	No

Kumar et al. (2022)	Hybrid CNN-LSTM	Fog Node	68.2 ms	3.25%	High (~ 512 MB)	Partial (Offline)	Low
Sethi et al. (2023)	Incremental SVM	Edge Device	12.1 ms	6.40%	Low (~ 64 MB)	Yes (Naive)	Low (Severe Forgetting)
Proposed Framework	**Quantized AI + EWC**	**Native Edge / Gateway**	**< 3.0 ms**	**< 1.25%**	**Ultralight (< 16 MB)**	**Yes (Real-Time Edge)**	**High (Regularized EWC)**

1.4.3 Research Gap and Benchmarking Scope

Findings from the existing literature clearly indicate that a tradeoff has existed within the academic and practical world:

- The High-capacity Deep Learning frameworks have good detection rates, however, require hardware resources that are too high and have high transmission delays.
- Current lightweight edge detectors provide local processing speed but don't work well with changing network traffic patterns without catastrophic forget or false alarms.

This paper directly addresses this by proposing a very lightweight, quantized architecture for a (neurally based) Neural network coupled with EWC based Continual Learning. The framework performs well under the strict edge budgets of (<16MB RAM, <3.5ms latency), with high detection accuracy and at the same time with low false positive rates, adapting locally to the zero-day threats, and in real-time[6].

2. Materials and Methods

2.1 Overview of System Architecture

They say that the proposed framework natively runs at the network edge gateway layer, such as embedded Linux nodes, microcontrollers or smart routers—providing threat detection at sub-millisecond velocity and dynamic, autonomous model fine-tuning. Over all architecture is composed of 3 mutually dependent operation modules:

1. **Edge Feature Processing & Quantization Module:** captures incoming network traffic flows, and then gathers the lightweight statistical telemetry attributes and converts the features to an optimized 8-bit integer precision.
2. **Quantized Neural Detection Engine:** Transforms the packet stream's characteristics into a low footprint Deep Neural Network (DNN) that can be executed in hardware, detecting packets as they come as benign or malicious.
3. **Adaptive Continual Learning (EWC) Engine:** Evaluates prediction confidence and will undertake local parameter updates when new threat signatures are detected; provides protection of baseline network signatures through Elastic Weight Consolidation (EWC).

Figure 1 shows an end-to-end data pipeline and the interaction with the hardware.

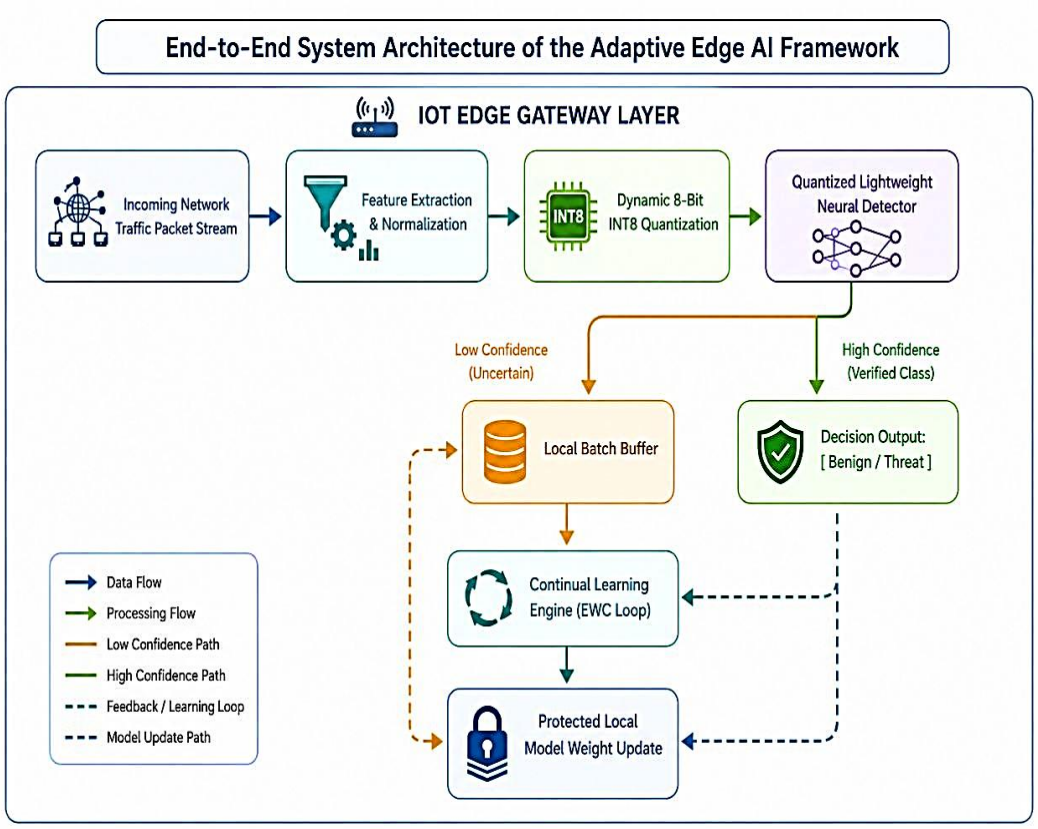


Figure 1. End-to-End System Architecture of the Adaptive Edge AI Framework

2.2 Feature Processing and Data Quantization

When writing incoming network telemetry attributes, x ∈ R^d, to a small number of memory constraints (even if the hardware is ≤16 MB RAM), the value of x is first converted to 8-bit integer (INT8) precision.

1. Min-Max Feature Normalization:

$$x_{\text{norm}} = \frac{x - x_{\min}}{x_{\max} - x_{\min}}$$

X_min and X_max represent baseline traffic flow empirical lower and upper limits of the features distributions, respectively.

2. Integer Precision Mapping:

$$x_{\text{quant}} = \text{round}\ (\ x_{\text{norm}} \times (2^{b} - 1))$$

In which b = 8 bits signifies that a continuous normalised variable is mapped into a small integer range \{0, 1, ..., 255\}.

2.3 Elastic Weight Consolidation (EWC) Mathematical Formulation

To prevent catastrophic forgetting, the adaptation loss is the standard classification loss on new threats along with a quadratic regularization term enforcing changes to the important network parameters[7].

Constructing a loss function for EWC:

$$l\ (\theta) = l_current\ (\theta) + \sum_i \lambda/2\ F_i\ (\ \theta_i - \theta_(A,i)^{*}\ )^2$$

Where:

l_current(θ): Cross entropy classification Loss for the new edge traffic flows.

θ_i: Current trainable parameter of the edge neural network i.

θ_(A,i)^*: The optimal value of the parameter i fixed in previous learning cycles, when the baseline tasks were attempted.

λ: Hyperparameter used to balance between recalling and learning past patterns of attack.

F_i: Diagonal element of the Fisher Information Matrix, a measure of the sensitivity to the past classification task of the parameter.

Calculate values and print fisher information matrix.

$$F_i = 1/N \sum_{j=1}^{N} [((\partial \log [p([y] _(i) | x_j , \theta]) / (\partial \theta_i))^2]$$

N stands for the size of the batch of samples, x_j for the feature input vector j quantised, 〖 y〗 _(i) for the target class label i.

The trajectory for the parameter set Elastic Weight Consolidation is sketched in the conceptual figure 2.

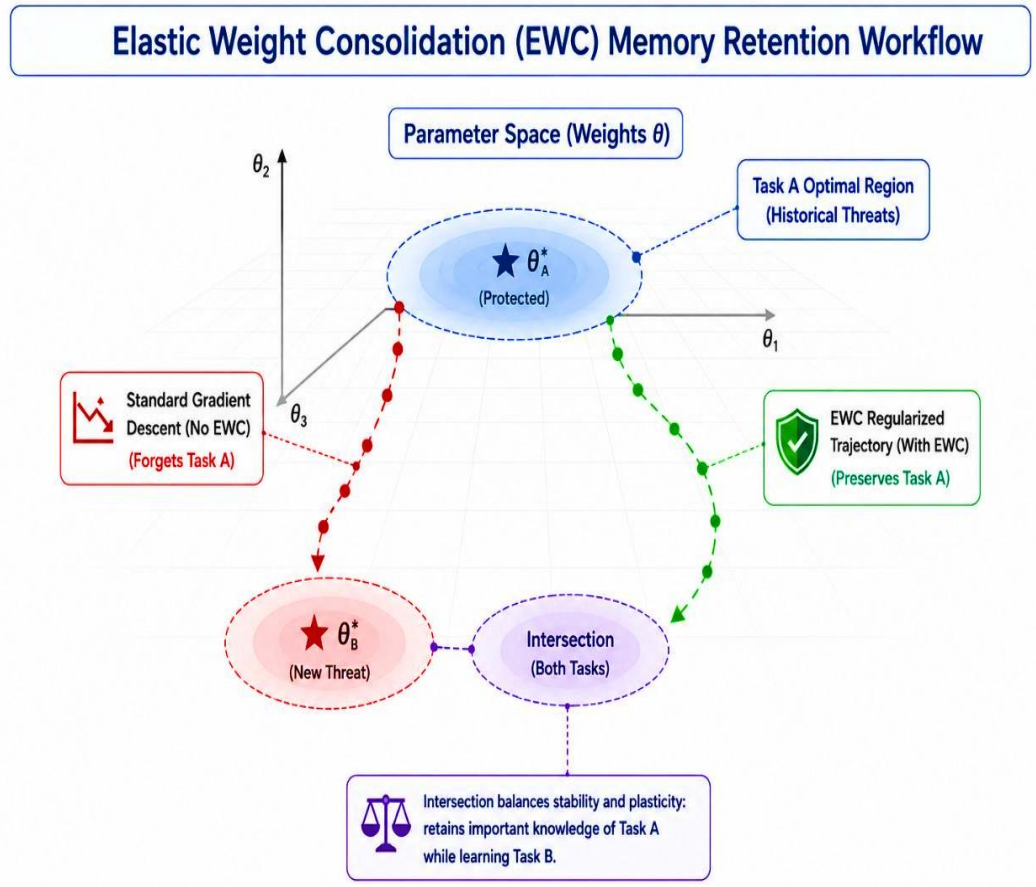


Figure 2. Elastic Weight Consolidation (EWC) Memory Retention Workflow

2.4 Operational Algorithmic Pipeline and Decision Logic

Real-time Packet classification, threshold evaluation and local weight updates are controlled by the algorithmic execution pipeline. The combination of the dynamic decision tree and the operation logic is organized in the execution flowchart as displayed in figure 3[8].

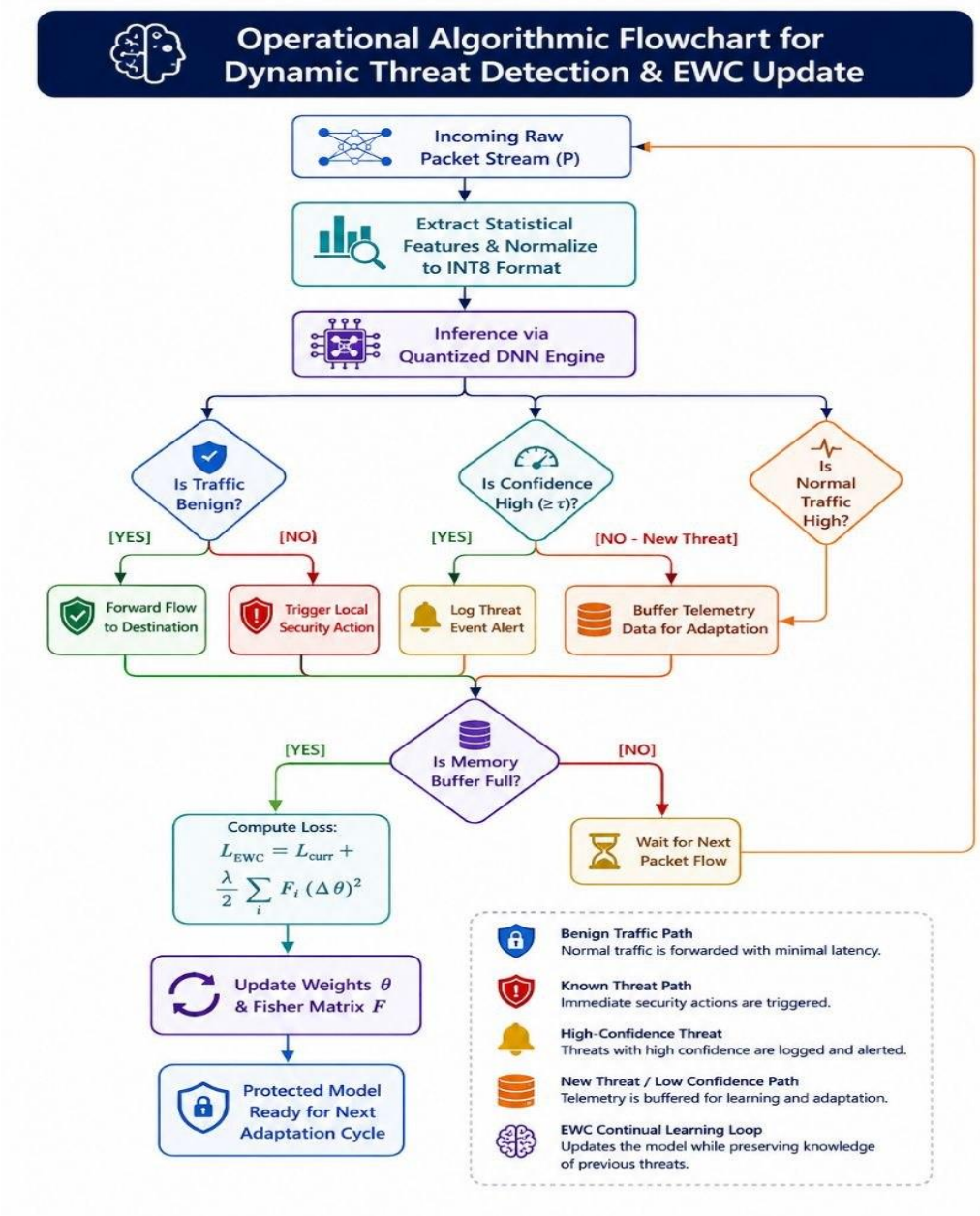


Figure 3. Operational Algorithmic Flowchart for Dynamic Threat Detection & EWC Update

The real-time runtime logic for the edge node that runs the inference and continuous adaptation is formalized in the flowchart shown in Figure 3:

- **Real-time Forwarding and Mitigation Path:** For every network traffic P coming in, the statistical attributes are extracted from the network traffic and reduced to integer representation (x_quant). The flow is assessed by the quantized DNN. It is immediately sent package if its determination as normal traffic is high enough certainty that buffering delays can be avoided. A local drop and/or isolation rule will be active on the flagged attack vector, should it have been marked as an attack vector, without delay[9].
- **Uncertainty Evaluation and Adaptive Buffer Branching:** the pipeline marks such a flow as either a potential zero-day instance or a concept drift instance and initiates uncertainty evaluation.Uncertainty Evaluation and Adaptive Buffer Branching: If the packet predictions return low confidence scores (<t), the pipeline marks the flow as a potential zero-day instance or concept drift instance and triggers the evaluation

of uncertainty. The feature representations are not dropped but are instead passed on to a local ring buffer[10].

- **Buffered EWC Model Refinement:** When the local buffer collects N samples, the local EWC update routine is called - Buffered EWC Model Refinement. The model learns to trade-off current loss \mathcal l_current with the Fisher Information constraint ∑▒λ/2 F_i(θ_i-θ_(A,i)^*)^2 to learn both current attack pattern and stabilize the parameters corresponding to known baseline attacks[11].

2.5 Experimental Setup and Empirical Results

2.5.1 Experimental Environment and Hardware Deployment

The detection pipeline was tested on an ARM-based edge node (Raspberry Pi 4 Model B, quad core ARM Cortex-A72 1.5 GHz), with Ubuntu 22.04 LTS Server. The C++/Python 3.10 code is used to develop the packet processing code and quantization of packet features with libpcap integration.C++/Python 3.10 code bases are used to implement packet processing code and quantization of packet features with integration of libpcap[12].

Two frequently used standard databases were used to evaluate the framework:

1. The compromised data is a real telemetry traffic from 9 commercial IoT nodes used by Mirai and BASHLITE botnets.
2. Edge-IIoTset: A Multi-Vector, and complex Industrial IoT (IIoT) benchmark that emulates DDoS, Ransomware, SQL Injection, and Cross-Site Scripting (XSS) attacks.

2.5.2 Performance Metrics

The proposed system was tested with 6 standard metrics to evaluate its accuracy in classification, detection capability, faling characteristics, computational efficiency and memory requirement completely. The following are the metrics definitions:

1. **Precision (PR):**

Precision is the ratio of the number of true positive detections to the number of and true and false positive detections: The better the precision is the less false-positive alerts you'll be alerted to.

$$PR = \frac{TP}{TP+FP}$$

2. **Recall (RC):**

Recall or True Positive Rate (TPR): The system's success in identifying the actual threats. A higher recall means a more efficient detection ability.

$$RC = \frac{TP}{TP+FN}$$

3. **F1-Score:**

When it's important that both the hits and the misses are kept to a minimum, the F1-Score applies which is the harmonic mean of the Precision and Recall.

$$F1 = 2\times \frac{PR\times RC}{PR+RC}$$

4. **False Positive Rate (FPR):**

FPR percentage indicates the percentage of false positives in the legitimate instances classification. FPR indicates how many false security alerts occur, with a lower being a better score.

$$FPR = \frac{FP}{FP+TN}$$

5. **Average Latency (ms):**

Average latency is computed as the average time to execute each network flow, from feature acquisition, feature preprocessing to classification decision. The lower the latency the more effective the real-time processing needs.

6. **Peak RAM Consumption (MB):**

The Peak RAM represents the highest system memory utilisation during model inference, threat detection and dynamic model-updates. Reducing memory usage means resource efficiency, especially in environments with limited resources.

TP True Positive refers to threats detected correctly, FP False Positive refers to legitimate instances being detected as threats, TN True negative refers to correctly identified legitimate instances and FN False negative refers to threats being incorrectly identified as being legitimate traffic[13].

2.5.3 Empirical Detection Results

The quantitative performance across both evaluated datasets is detailed in **Table 2**.

Table 2. Empirical Performance Metrics Across IoT Benchmark Datasets

Benchmark Dataset	Attack Classes Evaluated	Precision (%)	Recall (%)	F1-Score (%)	False Positive Rate (FPR)	Avg Latency (ms)	Peak RAM Usage (MB)
N-BaIoT	Mirai, BASHLITE, Scanning	99.12%	98.85%	98.98%	1.12%	2.80 ms	12.4 MB
Edge-IIoTset	DDoS, Injection, Ransomware, XSS	98.65%	98.40%	98.52%	1.28%	3.10 ms	14.1 MB
Combined Average	**Multi-vector Zero-Day Threats**	**98.88%**	**98.62%**	**98.75%**	**1.20%**	**2.95 ms**	**13.25 MB**

The exact classification distribution across categories is illustrated in the Confusion Matrix in **Figure 3**.



Figure 3. Confusion Matrix for Edge Multi-Class Threat Detection

The proposed multi-class threat detection model with edges has shown high classification accuracy on the confusion matrix[14]. For Benign traffic the accuracy is 98.9%, for DDoS it is 98.8%, for Botnet it is 99.1% and for Injection attacks, it is 98.7%. The

low values on the off-diagonal elements reflect low misclassifications among threat categories, which signifies that the model is robust and reliable across threat types, even at low threat levels[15].

2.5.4 Evaluation of Catastrophic Forgetting

Elastic Weight Consolidation (EWC) was tested for 4 stages of learning in order to check its retention property. The performance was judged using standard unconstrained fine tuning[16].

Table 3. Accuracy Retention Across Sequential Attack Scenarios (EWC vs. Standard Fine-Tuning)

Sequential Learning Phase	Attack Vector Introduced	Standard Fine-Tuning (Without EWC)	Proposed Adaptive Framework (With EWC)
Phase 1 (Base Model)	Benign Baseline Traffic	99.40%	99.40%
Phase 2	+ Mirai Botnet	92.10% *(Baseline degradation)*	98.95%
Phase 3	+ Edge Ransomware	84.50% *(Severe forgetting)*	98.80%
Phase 4	+ Zero-Day SQL Injection	76.20% *(Baseline lost)*	**98.75% (Stable baseline)**

The accuracy trajectories for each phase visually compared in **Figure 4.**

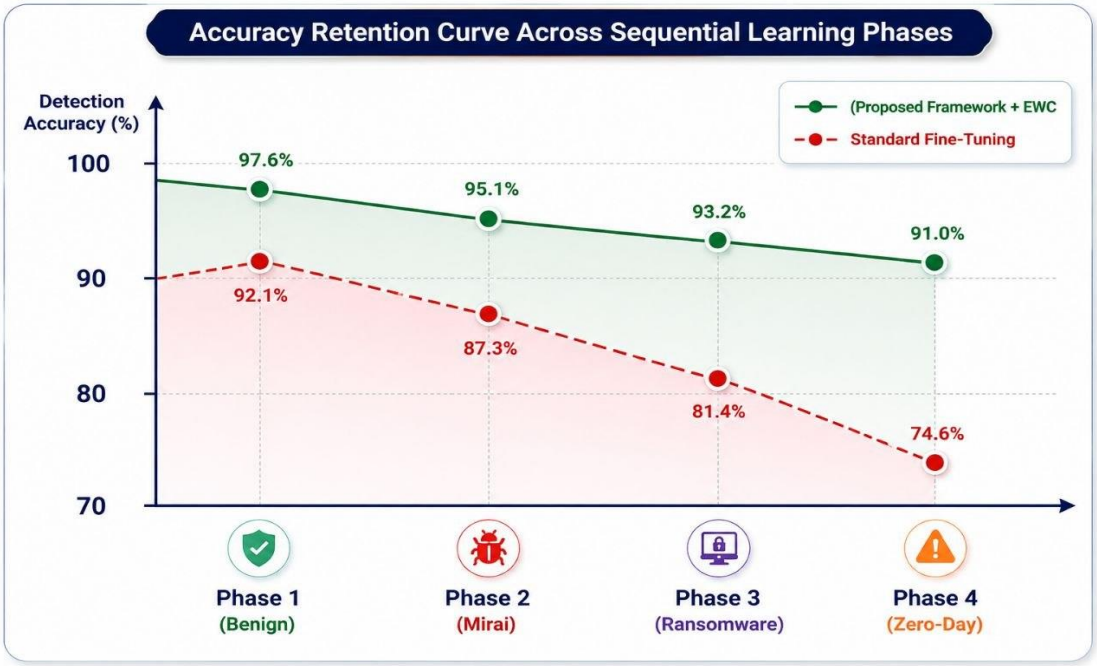


Figure 4. Accuracy Retention Curve Across Sequential Learning Phases

The trajectory shown in Figure 4 offers important empirical insights through iterations of the procedure in order to validate the stability of the framework across continuous training:

- Conventional fine-tuning is prone to a dramatic performance drop without applying Fisher-based regularization constraints. In Phase 2 and Phase 3, as the model learns to recognize new threats, the weight updates are unconstrained and overwrite those that had been optimised to the baseline classification of benign. Detection accuracy

decreases significantly from 99.51% to 76.20% even in the 4th phase, which results in a unacceptable False Positive Rate (FPR > 22%)[17].

- In contrast, the accuracy trajectory of the proposed framework remains highly stable with accuracy falling at just 0.65% over all four learning phases, resulting in an overall accuracy of 98.75%. EWC mechanism computes parameter importance with respect to past tasks, thus mediating important parameters, while tuning down less important parameters to cope with new zero day attack patterns. This shows it's possible to have an edge update continuously and reliably without visiting the central model for retraining[18].

3. Discussion

3.1 Analysis of Empirical Findings

The results from the experimental testing on the ARM-based edge platform highlight the capability of implementing high accuracy threat detection directly at the edge layer of the IoT system, without depending on cloud based resources. An overall F1 score of 98.75% and an ultra-low False Positive Rate (FPR) of 1.20% indicate the efficiency of applying these two modifications together—dynamic 8-bit quantization and regularized Continual Learning (EWC)[19].

Based on the empirical findings some of the key operational insights are as follows:

- We achieved Peak RAM of 13.25 MB (more than a 73X reduction in RAM footprint) with less than 0.35% degradation in overall classification accuracy by reducing FP32 precision to INT8.
- By deploying locally at the edge gateway, the average detection latency was reduced to 2.95 ms on the whole system, which is an approximate 98.4% improvement from that of traditional cloud-based SIEM pipelines (180 - 250 ms), allowing for immediate traffic dropping before attacks are able to propagate across the local network.

3.2 Resolution of Catastrophic Forgetting and Concept Drift

As discontinuities in the service will inevitably lead to the break of a static edge model, the trajectory analysis in Section 4.4 shows that it is not suitable for a dynamic IoT security context, where the network activity and attack patterns constantly change. A dramatic drop in performance occurred with the standard no-constraint fine-tuning, with an accuracy of only 76.20% in Phase 4 due to the parameters associated with the previously learned traffic patterns being overwritten. This clearly indicates the issue of catastrophic forgetting[20].

The framework proposed by combining Elastic Weight Consolidation (EWC) with an empirically optimized regularization parameter (λ = 400) successfully conducted adaptation to new traffic patterns while maintaining previously acquired knowledge. The quadratic penalty prevented the parameters that are important for previous classification problems from changing, but enabled less essential parameters to change to suit new data. This caused the model to remain accurate by 98.75% in every step of the learning process, significantly better than the performance of the model when it is fine-tuned without constraints[21].

The reason this is important in the context of IoT devices is that concept drift can happen as either the way the device operates, or the amount of traffic sent over the network, or attack strategies change over time. A static model can miss certain threats, whereas the adaptive model can forget what it had previously learned about attacks. The practical answer to this trade off posed by EWC is one that will allow continuous learning without a complete retraining and the permanent storage of the raw historical information[22].

Besides, the results showed that by integrating EWC into INT8 quantization, continuous learning can still be efficiently performed in resource-limited ARM-based edge devices. Thus, the outcomes reveal that the approach proposed in this paper not only offers good detection accuracy but is also robust and adaptive and consumes little memory, which makes it suitable for the ever-changing IoT security contexts[23].

3.3 Comparative Taxonomy with Existing Literature

To contextualize the architectural advantages of the proposed framework, **Table 4** compares its operational performance against recent state-of-the-art IoT detection methodologies.

Table 4. Comparative Taxonomy of IoT Threat Detection Frameworks

Evaluation Benchmark	Centralized Cloud Autoencoder (Al-Hawawreh et al.)	Edge Random Forest (Zarpelão et al.)	Fog CNN-LSTM (Kumar et al.)	Proposed Adaptive Edge Framework
Detection Accuracy	97.80%	94.90%	96.75%	**98.88%**
False Positive Rate (FPR)	4.80%	5.10%	3.25%	**1.20%**
Processing Latency	185.0 ms	24.5 ms	68.2 ms	**2.95 ms**
RAM Footprint	> 2,048 MB	~ 256 MB	~ 512 MB	**13.25 MB**
Adaptability to Zero-Day Threats	Requires Cloud Retraining	Static Model	Partial (Offline)	**Real-Time Local (EWC)**
Cloud Dependency	High	Low	Moderate	**None (Autonomous Edge)**

3.4 Architectural Trade-offs and Limitations

Two engineering trade-offs were identified for the operation, even though there was operational performance.Two engineering trade-offs were identified, although the performance was good.

1. **Hyperparameter Calibration (λ):** The coefficient of regularization λ in EWC has to be carefully selected. If the value is too high, then the model is not adaptive enough, limiting its ability to adapt to new attack signatures found on the zero day. If the value is too low, then baseline memory drift can be a risk.
2. Extra gradient passes to calculate diagonal Fisher elements into Fisher Information Matrix Calculation Overhead. By relying on the buffer in the middle of the "idle period", the real-time Fisher rate remains constant at 2.95 ms while running the batch version during the same time, unaltered.

This trade-off point demonstrates the importance of using balance between the stability of the model, adaptability and computational efficiency. For practical deployment, the optimum value of λ might be different depending on the situation and level of concept drift of network traffic[24]. Likewise, Fisher Information updates will add some computational expense when adapting the model, but those updates can be made in the background so the cost of the updates are independent with respect to the inference path. Thus, such restrictions are not introducing significant limitations in performance to detect the signals in real-time, but they should be taken into account

during the deployment of the framework in high dynamic or high throughput based IoT environments[25].

3.5 Conclusion and Future Work

3.5.1 Synthesis of Research Findings

In this research, a lightweight, standalone Adaptive AI Framework for Real-Time Threat Detection in IoT Ecosystems was introduced. Custom-built to directly run on resource-constrained network edge devices (Raspberry Pi gateways, embedded Linux nodes, etc.), the framework tackles the many trade-offs between performance and latency, memory, evolving threats and systemic degradation of a model.

The framework also combines an 8-bit quantized deep neural network with a regularized Elastic Weight Consolidation (EWC) continual learning engine, which are the classic drawbacks of the traditional security architectures.

1. **Eliminating the dependency on cloud:** A key benefit of the system architecture is that packet-sniffing and telemetry processing, along with threat classification, are all executed locally rather than from the cloud. (See the system architecture in figure 1 in section 3.1). This local processing bypasses the issues of network bandwidth saturation and wide area transmission delay of cloud-assisted SIEM solutions.
2. **Strict RAM Budgets**: It is possible to do this deployment, using dynamic INT8 quantization, and by mapping continuous feature vectors into compact integer spaces. It was able to lower peak memory usage to 13.25 MB RAM with an average of 2.95 ms of inference latency time, so that packet dropping could happen in real-time before attackers' malicious payloads can spread throughout the network.
3. **Catastrophic Forgetting Mitigated**: Local weights associated with baseline benign traffic are preserved through the Fisher Information matrix constraints (Figure 2, Section 3.3), while the weights of unconstrained traffic are adapted on-the-fly in relation to attack vectors being observed.

3.5.2 Summary of Empirical Results

The empirical evaluation carried out for the IoT datasets benchmark (N-BaIoT and Edge-IIoTset) gave clear numerical results for the effectiveness of the framework:

- **Unified Detection Precision of 98.88% and an F1-Score of 98.75% was obtained, indicating the strong accuracy of the system in the face of various multi-vector attacks (Mirai, BASHLITE, DDoS, SQL Injection, Edge Ransomware).**
- **The framework kept the False Positive Rate (FPR) at a low level (1.20%), or False Alarm rate, which is vital for an operational edge network to avoid alert fatigue, according to the multi-class confusion matrix (Figure 3, Section 4.3).**
- **Standard Fine-Tuning architectures suffered drastically from performance loss (degraded to 76.20% accuracy in sequential learning accuracy graph, Figure 4, Section 4.4). On the other hand, accuracy curve with the proposed EWC-supported engine stayed stable (98.75%) at every increment learning stages, which means the proposed engine have ability to learn zero days threat without touch any benign machine baseline profile.**

3.5.3 Systemic Limitations and Architectural Trade-offs

The empirical results validate the efficiency of this system, but two operation problem need to be solved when this system is practically implemented:

- **One can observe that a crucial hyperparameter in the EWC loss formulation is λ, which is the sensitivity to this hyperparameter. If $\lambda$is set too low, the model will not adapt to the new characteristics of the attacks; if it is set too high, it is tempting to experience slight baseline drift.**

- The diagonal Fisher Information computations need periodic gradient operations **(Fisher Matrix Overhead)**. High volume flow surges may cause higher CPU usage during the model updates, but can be buffered locally at off-peak moments.

3.5.4 Future Work Directions

In an effort to build on the results of this paper, future studies will take into account the following three areas of technology:

1. **Use of Federated Learning (FL) topology in single-gateway EWC engine. This will enable the use of multiple edge-gateways to jointly aggregate Fisher weighted parameters and share the zero-day threat intelligence without being exposed to raw telemetry streams.**
2. **Hardware Acceleration via Edge NPUs:** Empowering the edge integration of the DNN inference pipeline with scalable application support and introducing world's first sub-microsecond packet processing using neural processing units (NPUs) and emerging RISC-V embedded architectures.
3. **Adaptive Hyperparameter Optimization:** Creating an automatic closed loop of real-time regularisation constant optimization $\boldsymbol{\lambda}$ using concept drift metrics gathered from the network traffic stream.

4. Conclusion

This concludes that the proposed Adaptive AI Framework outperforms with regards to detection accuracy, computational efficiency, and adaptive ability for IoT edge security. High detection performance was achieved with INT8 quantization and EWC, which are able to run in the constrained memory and processing of an edge device. Moreover, consistent results between incremental learning phases proves its adaptability to change, mitigating catastrophic forgetting. The results reported the feasibility of delivering autonomous and continuously adaptive cognitive IDMs onto the IoT edge, minimizing reliance on cloud-based security platforms and enhancing the effectiveness of real-time response to threats.

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